Building a graduation project with **Gorse, Gin, Redis, and Typesense** is a sophisticated choice. Gorse is an AutoML-based engine, meaning it automatically tests and selects the best algorithms for your data.

Here is the breakdown of the specific algorithms used in this stack, what they do, and the industry-standard alternatives you can mention in your thesis.

---

1. The Core ML Algorithms (The "Brain")

Gorse primarily uses **Factorization Machines (FM)** and their derivatives for its "Ranking" stage.

Algorithm	**What it does**	**Alternatives (What else to use)**
Factorization Machines (FM)	Models interactions between features (e.g., "User Age" + "Post Topic"). Great for sparse data.	**DeepFM:** Adds a Neural Network to FM to capture complex, non-linear relationships.
BPR (Bayesian Personalized Ranking)	A "Pairwise" loss function. It learns that you prefer Post A over Post B, rather than giving them a 1-5 star rating.	**Pointwise (MSE):** Predicting an exact rating. **Listwise:** Optimizing the entire top 10 list at once.
Matrix Factorization (MF)	Decomposes the "User-Item" interaction matrix to find hidden (latent) patterns.	**SVD++:** A more advanced version that includes implicit feedback (views, not just likes).

2. Candidate Generation (The "Filter")

Since you can't run a complex AI on 1 million posts, these algorithms find the "top 1000" candidates first.

Technique	**Implementation in your stack**	**Alternatives**
Vector Search (ANN)	**Typesense** (using HNSW). Converts text/images into numbers and finds "neighbors."	**FAISS (Meta):** High-speed vector search for billions of items. **Milvus:** A cloud-native vector DB.

Technique	Implementation in your stack	Alternatives
Collaborative Filtering (CF)	**Gorse** finds "Users like you" or "Items similar to what you liked."	**LightGCN:** A Graph Neural Network approach to finding similar users.
Popularity-Based	**Redis** caches trending posts (most likes in the last hour).	**Thompson Sampling:** A bandit algorithm to balance "Trending" vs. "New" content.

3. Natural Language Processing (The "Context")

To understand what a post is actually about, you need an embedding model.

Tool	How to use it	Alternatives
BERT / RoBERTa	For converting social media text into vectors for **Typesense**.	**Llama 3 / Mistral:** Use an LLM to generate high-quality summaries of posts before embedding them.
Sentence-Transformers	The standard library for generating the numerical vectors.	**OpenAI Embeddings:** High accuracy but costs money per API call.

🚀 Comparison for your Documentation

In your final report, include a table like this to justify your choices:

Feature	Your Choice (Gorse/Go)	Traditional Choice (Python/Spark)	Why your choice is better?
Concurrency	**Go (Gin)**	Python (Flask)	Go handles thousands of simultaneous feed requests better.
Model Selection	**AutoML**	Manual Grid Search	Gorse automatically optimizes the model as more users join.

Feature	**Your Choice (Gorse/Go)**	**Traditional Choice (Python/Spark)**	**Why your choice is better?**
Search Speed	**Typesense**	Elasticsearch	Typesense is purpose-built for millisecond vector search.

Next Step for you:

Would you like me to help you write the **"Algorithm Selection"** section of your project proposal, explaining why Factorization Machines are superior to simple Matrix Factorization for a social media app?